Contents

1 Introduction

High quality elevation data is increasingly available across Europe through national and regional mapping programmes. These datasets support a wide range of applications, including hydrological modelling, flood risk assessment, infrastructure planning, and environmental management. At the same time, many of these applications extend beyond administrative boundaries. The European INSPIRE framework therefore places emphasis on the interoperability of spatial datasets and on their consistent use across national borders. Elevation is one of the spatial data themes addressed within this framework (INSPIRE, 2024).

The Rhine catchment is a clear example of an area where cross border spatial information is important. The catchment covers approximately 200,000 km² and extends across several European countries, with around 60 million people living within the basin. The Rhine and its tributaries are important for water supply, navigation, industry, agriculture, ecosystems, and flood management. Management of the river basin is already coordinated internationally through organisations such as the International Commission for the Protection of the Rhine (ICPR, 2022). Many processes within the catchment, particularly water flow and flooding, depend directly on terrain. The quality and consistency of elevation data can therefore influence analyses performed at catchment scale (Muthusamy et al., 2021).

Although detailed elevation datasets are available for large parts of Europe, they are generally collected and maintained by individual national or regional authorities. As a result, neighbouring datasets can differ in point density, spatial resolution, coordinate, and height reference systems, classification, and more. Recent work by EuroSDR has documented these differences across European point cloud datasets and has identified the absence of a harmonised cross border point cloud dataset at European scale (van der Heide et al., 2026). Such differences create practical problems when elevation data from several jurisdictions needs to be combined into a single terrain model. Simply joining the available datasets does not necessarily result in a consistent representation of the terrain.

This forms the main problem addressed by this project. The aim is to investigate how elevation data covering the Rhine catchment can be processed into a harmonised Digital Elevation Model in an automated manner. The project considers the differences between the available source datasets and the processing steps required to bring them into a common representation. The exact geographic coverage, datasets, and harmonisation aspects considered within the project are defined further through the research questions and subquestions.

This Project Initiation Document defines the approach that will be used to carry out the project. It establishes the research questions, scope, methodology, required data, planning, and expected outcomes. The following sections describe these elements in more detail and provide the basis for the execution and evaluation of the project.

2 Contributors

This section provides an overview of the student members of the team and their roles; as well as the clients and the supervision team.

2.1 Team Members

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Background:
Turkey
BSc Electrical Engineering, TU Eindhoven, The Netherlands

Interests: Point clouds, geospatial data visualization, digital terrain modelling

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Background:
Estonia
BSc Architecture, University of Bath, United Kingdom

Interests: Computational modelling of terrains and the built environment

Ruben Vons
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Background:
The Netherlands
BSc Aeronautical Engineering, Inholland, The Netherlands

Interests: Solving programming challenges and creating nice visuals.

2.2 Roles

TODO: Do we even want roles?

2.3 Supervision Team

The supervision team for this project consists of Gina Stavropoulou, representing TU Delft’s 3D Geoinformation Group, Maarten Pronk, representing Deltares, and Daan van der Heide, representing

Rijkswaterstaat; Maarten Pronk and Daan van der Heide are also PhD candidates in the 3D Geoinformation Group. The supervisors provide feedback on the team's work, help define scope and desired outcomes, and share their experience with existing approaches to the problem.

2.4 Clients

Rijkswaterstaat are a government agency for infrastructure and water management of the Netherlands. Originally created in 1798 for flood prevention, it now also oversees the construction and maintenance of national infrastructure (Rijkswaterstaat, n.d.). Some of the notable projects completed by the agency include the Afsluitdijk, built in the interwar period, and the Delta Works, the construction of which finished in 1997. Flooding still remains a key priority for Rijkswaterstaat.

Deltares are an independent research institute specialising in water and subsurface research (Deltares, n.d.). Their key mission is 'Enabling Delta Life', which encompasses five key areas focusing on health, safety, and sustainability of life in river deltas. Creating a more accurate river model contributes to several of these areas — in particular, "Safer from flooding" and "Healthy water systems".

3 Project Definition

3.1 Research Questions

The main research question of this project is:

How to create a harmonised cross-border DEM of the Rhine catchment area in an automated way?

As described in the introduction, the project focusses on the issues presented by the cross-border data of the Rhine catchment area. This translates to the creation of an automated pipeline and the report describing the issues that come with attempting the creation of a harmonised DEM. The efficiency refers to creating a pipeline that accounts for the large data quantity and limited computational resources. These two aspects of the main question can be broken down into the following subquestions:

- What is the Rhine catchment area?
- What, if any, existing approaches are there to creating a harmonised DEM from heterogeneous sources?
- What are the existing DEMs at different scales (regional, national, and global)?
- What are the issues with cross-border data?
- What are the INSPIRE requirements for DEMs?
- What are the limitations of an automated harmonisation pipeline?
- How can we test the accuracy of the reconstructed cross-border DEM?

Answering these subquestions will help to answer the main research question. The first five subquestions will be evaluated in the research phase of the project and relate mostly to existing works and available data. Answering these questions will inform data acquisition and subsequently, software development. The next research questions will be answered during the software development phase as they relate to testing and limitations of the pipeline. These questions will be answered by evaluating the procedures and the output, and discussing the findings with the client.

3.2 Relevant courses from the MSc Geomatics program

Do we care about this?

This project applies knowledge gained from several courses from the MSc Geomatics program. Having a basic overview of these courses will highlight the current knowledge possessed by the team. These are the following courses:

- GEO1000 - Python for Geomatics;
- GEO1001 - Sensing Technologies;
- GEO1002 - GIS and Cartography;
- GEO1004 - 3D modelling for the build environment;
- GEO1015 - Digital Terrain Modelling.

GEO1000 is relevant for programming the pipeline in either Python or C++. GEO1001 was the basis for understanding point cloud data collection and processing. GEO1002 is relevant for understanding the data and how to visualise it. GEO1004 is possibly relevant for working with 3D data and point cloud processing. GEO1015 is applicable as it forms the basis for 2D and 2.5D terrain modelling, shortcomings and processing of DEMs.

3.3 Requirements

This chapter describes the requirements for this project. The requirements are split into 5 categories: “DT” (for “data”) describes the datasets that are going to be used as input; “CT” (meaning “country”) describes the expected spatial extent of the output dataset within the Rhine catchment area;

“MP” (for “map”) describes the contents of the output data; “TC” refers to the technical requirements of the workflow; and “RP” stands for “report”. The requirements each have a priority assigned based on the MoSCoW method. This method divides requirements into four categories: “Must have”, “Should have”, “Could have”, and “Will not have”. “Must have” requirements are mandatory; “Should have” requirements are not mandatory but nice-to-have or would add value to the project; “Could have” requirements are have been discussed but add limited value or require more work. “Will not have” features have been determined to be strictly outside of scope of the project as impractical due to high labour, data, or technical cost. The full assessment is shown in Table 1.

Req ID	Description	MoSCoW
DT-01	Global/EU DEM	Must
DT-02	Rhine Watershed Mask	Must
DT-03	Rhine Bathymetry	Will not have
CT-01	Netherlands is included	Must
CT-02	Germany is included	Must
CT-03	Belgium is included	Should
CT-04	Switzerland is included	Must
CT-05	France is included	Must
CT-06	Luxembourg is included	Must
CT-07	Austria is included	Must
CT-08	Liechtenstein is included	Must
CT-09	Italy is included	Will not have
MP-01	DSM map	Must
MP-02	DTM map	Must
MP-03	Land-sea mask	Must
MP-04	Bathymetry mask	Could
MP-05	Nodata mask	Should
MP-06	Point cloud density map	Should
TC-01	Workflow is fully automated	Must
TC-02	Workflow is efficient	Should
TC-03	Workflow is reproducible	Must
RP-01	Report on findings	Must
RP-02	Report details the workflow	Must

Req ID	Description	MoSCoW
RP-03	Report details the issues with cross-border data	Must
RP-04	Report details the limitations of the workflow	Must

Table 1: MoSCoW prioritization of data requirements for the project

4 Methodology

This chapter describes the expected approach to research and development that will be performed over the course of the project. In addition, methods for quality assurance are discussed.

4.1 Research Approach

4.2 Pipeline Design

Based on the requirements outlined in Table 1, it is anticipated that the technical pipeline will consist of the following steps:

1. Data acquisition
2. Data preprocessing
3. Point cloud processing
4. Raster DEM processing
5. Data postprocessing
6. Documentation

Below is the detailed description of each of the steps above.

4.2.1 Data Discovery and Acquisition

Data acquisition chiefly refers to getting access to the necessary datasets — global, national, and regional — and downloading them. For each country, the data should include at minimum a point cloud, a DEM (ideally, separate DSM and DTM); and if possible, also a topographic dataset describing bodies of water. Most of this data is covered under INSPIRE and therefore should be available from national geoportals. The full list of datasets is found in Appendix A.

4.2.2 Data Preprocessing

At this step, the spatial extent and other characteristics of input data and the resolution of rasters is determined — this resolution in particular informs the resolution of the output. The raster files are cropped to the spatial extent of the Rhine catchment area; and the point clouds are reduced only to the border areas where there is intersection and potential conflict in height data. It is expected that this step is carried out using Python libraries such as `PDAL`, `GDAL`, `rasterio`, and `laspy`.

4.2.3 Point Cloud Processing

This stage of the pipeline includes:

- converting the point clouds to the same CRS;
- aligning them if the CRS conversion is determined to be insufficiently accurate;
- performing the ground filtering to extract the DTM;
- rasterising the output.

While the majority of these operations are supported by `PDAL` and `rasterio`, we anticipate that aligning heterogeneous point clouds is going to require additional research into tools and subsequent development and testing.

4.2.4 Raster Processing

This stage of the pipeline includes:

- converting raster files to the same CRS;
- if necessary, further aligning them using the same transformation as the respective point cloud;
- raster resampling to align their resolution;
- interpolating if a raster image has gaps (such as in the case of the Netherlands' DTM);
- stitching the individual tiles into one file.

At the end of this stage, the raster DEM is complete.

4.2.5 Data Postprocessing

To make the raster usable and for broader visualisation purposes, some further postprocessing may be required. In Table 1, this is referred to under a blanket term of "auxiliary masks and rasters", which may include the following:

- a land/sea mask;
- a land/inland water mask;
- a "NoData" mask (i.e. regions of NoData in the original raster file);
- a point density map;
- "Border" mask (i.e. regions where heterogeneous datasets had to be harmonised).

4.2.6 Documentation

Per the clients' request, the software is also going to be documented. The documentation will cover how to run the software (essentially a `README`) but also the challenges, issues, and limitations that the team will have encountered over the course of the project. This documentation will be separate from, but partly overlapping with, the report.

4.3 Quality Assurance

We anticipate several aspects to quality assurance:

- **Evaluating input data.** This includes examining the existing DEMs for resolution, coverage, interpolation approaches, and whether both DSM and DTM are provided; and PCs for density, coverage, and classification.
- **Test runs to evaluate pipeline.** Using both synthetic and smaller samples of real-world data, evaluate the harmonisation of the point cloud and subsequently, the DEM.
- **INSPIRE compliance.** The output should be compliant with the INSPIRE Data Specification on Elevation. This is to be ensured both at the development stage and by evaluating samples of the output.
- **Comparison to existing DEMs.** The best candidate for this step is Copernicus DEM published by the European Space Agency. While we anticipate both spatial (mainly resolution — Copernicus DEM is freely available at 30m resolution while expected resolution of Rhine DEM is 5m or finer) and temporal (dates of acquisition) discrepancies, this should still provide some overview of the pipeline's performance across different areas of the catchment area.

Furthermore, clients will provide their input on our progress so further quality assurance steps will be implemented if necessary.

5 Planning

5.1 Project Phases

5.1.1 Organisation phase

The organisation phase is the first phase of the project. During this phase, the project will be defined and planned. This phase will be used to define the project scope, objectives, deliverables, and timeline. The project team will be formed and roles and responsibilities will be assigned. The project plan will be created and approved by the stakeholders. This phase will last approximately 2 weeks.

The phase consists of the following tasks:
- Project Initiation Document (PID) creation and approval;
- Project planning and scheduling;
- Meeting with stakeholders to discuss project scope, objectives, deliverables, and timeline.

This report is the PID and includes most information aquired during this first phase. The project plan and schedule can be found at the end of this chapter in the form of a Gantt chart. The meeting minutes can be found on the project repository. While elements discussed in this first phase are not final, they will be used as a basis for the project plan and schedule.

5.1.2 Research phase

The research phase is the second phase of the project. During this phase, existing approaches and methods will be evaluated. Additionally, during this phase, data will be collected and evaluated for fitness of use and quality. It is expected that this phase will last approximately 2 weeks.

The phase consists of the following tasks:
- Literature review of existing approaches and methods;
- Review of CRS transformations and their accuracy;
- Refinement of the methodology and approach based on the literature review;
- Data collection and evaluation for fitness of use and quality.

It is likely that other projects have developed different approaches to the problem of creating a cross-border DEM. This will be the main focus of the literature study. Additionally, since each country has its data in a different CRS, it is important to learn how to transform the data into a common CRS. This will be done by reviewing CRS transformations and their accuracy. The methodology and approach described in this PID is based on an intial assumption of the problem and will be refined based on our findings. Lastly, data for both the ground truth and the point clouds will be collected and evaluated in preparation for the next phase.

5.1.3 Software development and data collection phase

The software development and data collection phase is the third phase of the project. During this phase, the software will be developed and tested. This includes both a pipeline prototype and a production version. It is expected that this phase will last approximately 4 weeks.

The phase consists of the following tasks:

- Development of a pipeline prototype;
- Testing of the pipeline prototype on a data subset;
- Development of a production version of the pipeline;
- Testing of the production version of the pipeline a data subset;
- Testing of the production version of the pipeline on the full dataset;
- Software documentation and user manual creation.

The pipeline prototype will be developed as a proof of concept and will be test on a small data subset. For the production version, the pipeline will be developed to handle the full dataset. The production version will be tested on a small data subset and then on the full dataset. The software documentation and user manual will be created to ensure that the software can be used by others.

5.1.4 Midterm presentation phase

The midterm presentation phase is the fourth phase, but will be executed in parallel with the software development and data collection phase. During this phase, the project will be presented to the stakeholders to show the progress made and to receive feedback. In addition, this opertunity may be used to adjust the project scope or objectives. There is no set duration for this phase.

The phase consists of the following tasks:

- Midterm presentation preparation;
- Midterm report creation.

The midterm presentation will be prepared to show the progress made up until that point. The midterm report will be the progress of the final report and will contain the results of the research phase and the software development in its current state. Both the presentation and report will be presented to the stakeholders to receive feedback and to adjust the project scope and/or objectives if neccesary.

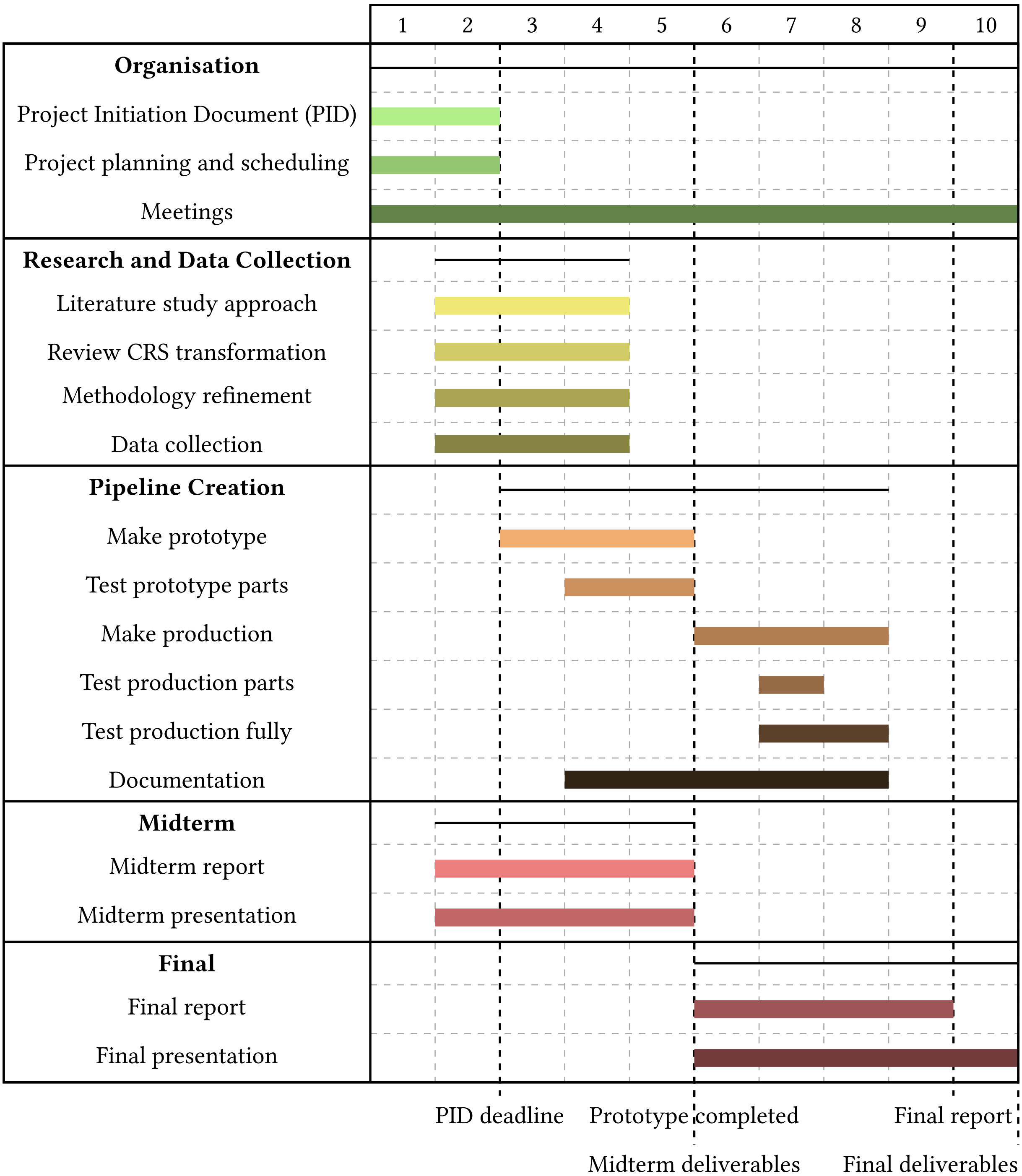
5.1.5 Final presentation phase

The final presentation phase is the fifth and final phase of the project. During this phase, the project will be presented to the stakeholders to show the final results and the report. In addition, this will contain the final geomatics day presentation. There is no set duration for this phase.

The phase consists of the following tasks:

- Final presentation preparation;
- Final report creation.

The final presentation will be prepared to show the final results and conclusions of the project. The final report will contain the results of the research phase and the software development in its final state.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Organisation
Project Initiation Document (PID)
Project planning and scheduling
Meetings
Research and Data Collection
Literature study approach
Review CRS transformation
Methodology refinement
Data collection
Pipeline Creation
Make prototype
Test prototype parts
Make production
Test production parts
Test production fully
Documentation
Midterm
Midterm report
Midterm presentation
Final
Final report
Final presentation
PID deadline
Prototype completed
Final report
Midterm deliverables
Final deliverables

5.2 Communication

Communication is key to the success of this project. To ensure communication is maintained, both within the team and with stakeholders, regular meetings will be held. The team will meet on a weekly basis to discuss progress, issues and questions. These meetings will be held on Fridays. There will be an additional opertunity to meet on Mondays if neccesary. For each meeting, key talking points will be shared in advance. Meeting minutes will be kept and distributed after each meeting. These meetings will be held online via Microsoft Teams. The team will also communicate internally outside of these meetings, either through Whatsapp or in person.

5.3 Risk Analysis

In order to identify and mitigate risks, a risk analysis will be conducted. The method used is the risk matrix method. It is important to mention that this method is not infalible and that it is possible that risks are not identified or that the impact and likelihood are wrongly assessed. However, this method does allow for a more structured approach to risk analysis. Table 2 shows the identified risks, their impact and likelihood, and what can be done to mitigate them.

Likelihood	Harm severity			
	Minor	Marginal	Critical	Catastrophic
Certain	High	High	Very high	Very high
Likely	Medium	High	High	Very high
Possible	Low	Medium	High	Very high
Unlikely	Low	Medium	Medium	High
Rare	Low	Low	Medium	Medium
Eliminated	Eliminated			

Figure 1: Risk Assessment Matrix.

Risk ID	Description	Impact	Likelihood	Mitigation
1.	Pipeline too computationally taxing or insufficient computational resources.	Critical	Possible	Test with small dataset and adjust spatial extent if necessary.
2.	Insufficient quality/ availability of point cloud data.	Marginal	Rare	Shift focus to regions with available high-quality data.
3.	CRS transformations are inaccurate.	Critical	Unlikely	Test CRS alignment on select border regions
4.	Missing/conflicting data on border areas.	Critical	Likely	Check spatial overlap of datasets and prioritize one dataset.
5.	Task is too ambitious given timeframe/ team size.	Catastrophic	Unlikely	Check if internal deadlines are met

Risk ID	Description	Impact	Likelihood	Mitigation
				and adjust scope if necessary.
6.	Edge cases not properly assessed due to data quantity and variability.	Minor	Possible	Unideal but acceptable within scope of project.
7.	Task is insufficiently constrained/defined.	Critical	Unlikely	Regular meetings with team and stakeholders to assess and adjust scope and requirements if necessary.
8.	Lacking/inadequate communication within the team and/ or with stakeholders.	Critical	Possible	Schedule regular meetings and maintain open communication channels.
9.	Discovery of unexpected issues.	Marginal	Possible	Regular meetings with team and stakeholders to assess impact and devise mitigation strategies if necessary.
10.	Temporary or permanent absence of team members.	Critical	Rare	Set internal deadlines early to allow for adjustments in case of absence.
11.	Data loss.	Catastrophic	Possible	Implement regular backup procedures.

Table 2: Risk Assessment Table.

https://en.wikipedia.org/wiki/Risk_matrix

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A Data Sources